

Group 5

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Quotient Group

Claim 1. *If a subgroup H of a group G is not normal, there are left cosets aH and bH whose product is not a coset.*

Proof. Let $H \subset G$ be a subgroup of G which is not normal. Assume the contrary that the product of left cosets aH and bH , abH is always a coset. $\forall g \in G$, let $a = g$ and $b = g^{-1} \in G$. Thus $abH = eH = H$. By definition of a coset $gh \in gH$ for $h \in H$ by definition of a coset. Thus $ghg^{-1} \in abH = H$ by definition of a coset. Thus $ghg^{-1} \in H$ for all $g \in G$. Thus H is normal. Thus a contradiction forms and there exists left cosets aH and bH whose product is not a coset. $\square$

Problem 2. *In the general linear group $GL_3(\mathbb{R})$, consider the subsets*

$$H = \begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

, and

$$K = \begin{bmatrix} 1 & 0 & * \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where $*$ represents an arbitrary real number. Show that H is a subgroup, that K is a normal subgroup of H , and identify the quotient group H/K . Determine the center of H

Solution. H is closed because the product of 2 upper triangular matrix is upper triangular as well. $I \in H$, thus H has the identity. Finally $\forall A \in H$, $A = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix} \in H$. Thus H is a subgroup.

$K \subset H$ containing the identity, and elements of K can be multiplied by adding the upper-right elements together, implying that K satisfies the conditions to be a subgroup of H . Furthermore multiplication between elements of K and elements of H is given by, $\forall A \in K$ and $\forall B \in H$

$$AB = \begin{bmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a & b+x \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a & b+x \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$$

Thus $\forall A \in K$ and $\forall B \in H$, $AB = BA$ thus $K \subset Z(H)$ and K is normal.

$\forall h \in H$, the coset hk would contain all elements of H that are equal to h in the 1, 2 and 3 entries.

Call this coset C_{ac} where $h_{1,2} = a$ and $h_{2,3} = c$. Lets see how the cosets interact

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & d+a & e+af+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{bmatrix} \in C_{d+af+c}$$

Thus the product of C_{ac} and C_{de} is C_{d+af+c} so the quotient group is isomorphic to $\mathbb{R}^+ \times \mathbb{R}^+$. Let

$$\forall A \in H, A = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \text{ and } \forall D \in H, D = \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix}$$

$$AD = \begin{bmatrix} 1 & d+a & e+af+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } DA = \begin{bmatrix} 1 & d+a & e+cd+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{bmatrix}$$

Thus if $AD = DA$, $af = cd$. If either a or c is nonzero, it is always possible to choose f and d such that $af \neq cd$, thus the center of H is K . ■

Claim 3. Let P be a partition of a group G with the property that for any pair of elements A, B of the partition, the product set AB is contained entirely within another element C of the partition. Let N be the element of P that contains 1. Prove that N is a normal subgroup of G and that P is the set of its cosets.

Proof. Let P be a partition of a group G with the property that for any pair of elements A, B of the partition, the product set AB is contained entirely within another element C of the partition. If $AB \in C$ and if $AB \in C'$ then $C = C'$ because the elements of a partition are pairwise disjoint.

Define a multiplication on the partition such that $A * B = C$ s.t $AB \subset C$ s.t $C \in P$. Such a unique C will always exist by the definition of our partition, thus the multiplication is well defined. Furthermore by the equivalence relation homework, we know a partition and the set of all equivalence classes for a given set are the same thing. Then define the following $\sim \forall a, b \in G, a \sim b \iff a, b$ in the same element of the partition P . Thus if $a \in A$ then $[a] = A$. Then for $a \in A$ and $b \in B$, the product set $AB = [a][b] \subset C$, however by definition of product set $ab \in C$ thus $[ab] = C$ and $[a][b] \subset [c]$ and $[a] * [b] = [c]$. Create a natural map $f : G \twoheadrightarrow G/\sim$ such that $f(a) = [a]$. Let $N = [e]$. Then for all $a \in G$, $[e][a] = [a]$ because $e \in [e]$ and $a \in [a]$, thus $ea = a \in C$ but since C is a element of P containing a then $C = [a]$, thus $[e][a] \in [a]$ and $[e] * [a] = [a]$. The argument is symmetric for $[a] * [e] = [a]$ thus $[e]$ is the identity in $G/\sim$. Furthermore, $\forall g \in G, [g] * [g^{-1}] = [gg^{-1}] = [e]$ thus $[g]^{-1} = [g^{-1}]$ similarly, $[g^{-1}][g] = [g^{-1}g] = [e]$. Thus $G/\sim$ has inverses. Finally $\forall a, b, c \in G, [a] * ([b] * [c]) = [a] * [bc] = [abc] = f(abc)$ and $([a] * [b]) * [c] = [ab] * [c] = *[abc] = f(abc)$. Thus $G/\sim$ is associative. Thus $G/\sim$ is a group. Finally $\forall a, b \in G, f(ab) = [ab] = [a] * [b] = f(a) * f(b)$ thus f is a homomorphism. Thus $[e] = N$ is the kernel of the homomorphism f and thus must be normal. $G/\sim \cong P$ is the set of its cosets. □

Problem 4. Let $H = \{\pm 1, \pm i\}$ be a subgroup of $G = C^\times$ of the fourth roots of unity. Describe the cosets of H in G explicitly. Is $G/H \cong G$?

Proof. $\forall g \in G, gHg^{-1} = gg^{-1}H$ because G is abelian and $gHg^{-1} = H$ and thus H is normal.

For all $g \in G, gH = \{g, -g, gi, -gi\}$. There exists a surjective homomorphism $f : C^\times \rightarrow C^\times$ which takes $x \rightarrow x^4$. H is the kernel of f , because $f(x) = 1 \iff x \in [\pm 1, \pm i] = H$. Furthermore there exists a natural surjective homomorphism from $p : C^\times \rightarrow C^\times/H$. Thus since $\ker p = H \subseteq \ker f = H$, f factors through p through the prehomework theorem. Thus the homomorphism F such that $f = F \circ p$ is surjective because p and f are surjective. Furthermore $\forall aH, bH \in C^\times/H$, if $F(aH) = F(bH)$, then $f(a) = f(b)$ and $f(ab^{-1}) = 1$ thus $ab^{-1} \in H$. Since $C^\times$ is abelian, $ab^{-1} = b^{-1}a \in H$. By definition of a coset, $bb^{-1}a \in bH$ thus $a \in bH$ however since H is normal cosets are pairwise disjoint and thus $aH = bH$. Thus F is injective. thus there exists a bijection and a homomorphism from $G/H \rightarrow C^\times$ and thus $G/H \cong C^\times$. $\square$

Problem 5. Let G be the group of upper triangular real matrices $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$, with a and b different from zero. For each of the following subsets, determine whether or not S is a subgroup, and whether or not S is a normal subgroup. If S is a normal subgroup, identify the quotient group G/S .

Solution. 1. S is the subset defined by $b = 0$. S is a subgroup because two diagonal matrices multiply result in a diagonal matrix, I is in here, and inverses exist as well. S is not normal because

$$\begin{bmatrix} x & y \\ 0 & z \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} 1 & \\ & ad \end{bmatrix} \begin{bmatrix} z & -y \\ 0 & x \end{bmatrix} = \frac{1}{ad} \begin{bmatrix} axz & xy(d-a) \\ 0 & dzx \end{bmatrix} \notin S$$

2. S is the subset $d = 1$. S is not a subgroup because $\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}^{-1} = \frac{1}{a} \begin{bmatrix} 1 & -b \\ 0 & a \end{bmatrix} \notin S$ if $a \neq 1$.

3. S is a subset $a = d$

S is closed $\forall A, X \in S, AX = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \begin{bmatrix} x & y \\ 0 & x \end{bmatrix} = \begin{bmatrix} ax & ax+by \\ 0 & ax \end{bmatrix} \in S$. S contains the identity and $\forall A \in S, A^{-1} = \frac{1}{a^2} \begin{bmatrix} a & -b \\ 0 & a \end{bmatrix} \in S$. Thus S is a subgroup.

$\forall X \in G$ and $\forall A \in S, XAX^{-1} = \frac{1}{xz} \begin{bmatrix} x & y \\ 0 & z \end{bmatrix} \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \begin{bmatrix} z & y^{-1} \\ 0 & x \end{bmatrix} = \frac{1}{xz} \begin{bmatrix} axz & bx^2 \\ 0 & axz \end{bmatrix} \in S$. Thus S is normal.

The quotient group G/S would be represented by matrices of the form

$$\begin{bmatrix} 1 & 0 \\ 0 & d \end{bmatrix},$$

for $d \in \mathbb{R}^\times$ since this would cover all of G :

$$\begin{bmatrix} 1 & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & ad \end{bmatrix},$$

and since each matrix of the form above gives a different coset since multiplying by elements in S cannot give another matrix of the same form unless d is the same. $\blacksquare$

Miscellaneous

Problem 1. Describe the column vectors $(a, c)^T$ that occur as the first column of an integer matrix A whose inverse is also an integer matrix.

Proof. A is a 2×2 matrix where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a, b, c, d \in \mathbb{Z}$. $\det(I) = \det(AA^{-1}) = \det(A) \det(A^{-1}) =$

1. Since A, A^{-1} has integer entries $\det(A), \det(A^{-1}) \in \mathbb{Z}$. Thus $\det(A) = 1$. Thus $ad - bc = 1$. Let $b' = -b$, thus $ad + b'c = 1$. Thus there exists $b', d \in \mathbb{Z}$, such that $ad + b'c = 1$, thus by Bezout's theorem, $\gcd(a, c) = 1$. If $a = c = 0$, then $\det(A) = 0$ and it is singular thus forming a contradiction. If $\gcd(a, c) > 1$, then by Bezout's theorem the determinant of any integer matrix with this property will also be a multiple of that GCD. So it is only the coprime a, c . $\square$

Claim 2a. Every group of even order contains a element of order 2.

Proof. Let G be a group with even order. Define the equivalence relation, $\forall a, b \in G \ a \sim b \iff b = a^{-1}$. The number of equivalence classes of odd size must be even. The element e is in its own equivalence class, so there must be some equivalence class of odd size. Moreover since elements can only have 1 unique inverse the equivalence classes can only contain 1 element (if the element in the equivalence class is its own inverse) or 2 (element and its inverse). So there must be a $a \in G$ s.t $a = a^{-1}$ $\square$

Claim 2b. Every group of order 21 contains a element of order 3

Proof. Let G be a group with order 21. By Lagrange's Theorem, the order of cyclic subgroups of G must be 21, 7, 3, 1.

1. If $S \subseteq G$ has order 21 and is cyclic, then $S = G$ and S is generated by a single element $g \in G$ s.t $g^{21} = e$, then $g^7 = (g^7)^3 = e$. Let $g^7 = a$, thus $a^3 = e$ and order of a is 3 and thus there exists a element of order 3.
2. If all cyclic subgroups have order 1 they are all e thus $|G| \neq 7$ and thus there must be a cyclic subgroup of different order.
3. Cyclic subgroup of order 3 is trivial because the element generating the subgroup will have order 3 and thus G contains a element of order 3.
4. Suppose all cyclic subgroups of G have order 7. Let $g, h \in G$, suppose $\exists f \in \langle g \rangle \cap \langle h \rangle$, then $f = g^a$ and $f = h^b$ for $a, b \in [0, \dots, 6]$. If $a = b = 0$, then e would be in both subgroups. However if $a \neq 0$ and $b \neq 0$, thus $g^a = f^b$ but since 7 is prime and thus a coprime to 7, thus $g^a = f^b$ generates $\langle g \rangle$ but since b also coprime to 7, $g^a = f^b$ generates $\langle f \rangle$. Thus $f \in \langle g \rangle$ thus $\langle g \rangle = \langle f \rangle$. Thus the only element in the intersection of the cyclic subgroups with order 7 is e and all the other 6 elements in each subgroup were not in the intersection with any other cyclic subgroup. Thus the number of elements is $6n + 1$ where n is the number of subgroups. However there exists no $n \in \mathbb{Z}$ s.t $21 = 6n + 1$ because $6 \nmid 20$. Thus a contradiction forms and there cannot be cyclic groups of order 7 only and thus there must exist a cyclic group of order 3 and thus a element of order tree.

$\square$

Problem 3. *Classify cyclic groups of order 6*

Claim 3a. *Let G be a group in which every element has order 2. Then G is Abelian*

Proof. Let G be a group, in which every element has order 2. Thus $\forall a \in G, a^2 = e$ thus $a = a^{-1}$. Thus for all $a, b \in G, ab = a^{-1}b^{-1} = (ba)^{-1} = ba$. Thus G is abelian $\square$

Claim 3b. *Let G be a group in which every element has order 2, $G \cong F_2^n$*

Proof. Since G is abelian, let it be additive thus $(G, +) \cong G$. Thus $\forall x \in (G, +), x + x = 2x = 0$. Thus $\forall x \in (G, +), ax = (a \bmod 2)x$. Vectorspaces under F_2 are also abelian and have this property and have all other properties of $(G, +)$. Thus $(G, +)$ is isomorphic to Vectorspaces under F_2 . Assuming G is finite, define $f : (G, +) \rightarrow F_2^n$. Let $G = \{0, x_1, \dots, x_n\}$ be the unique elements of G . $f(x_i) = e_i^n$ and $f(0) = 0$. $f(x_i + x_j) = e_i + e_j = f(x_i) + f(x_j)$ thus f is a homomorphism. f is an isomorphism and thus $G \cong F_2^n$. $\square$

Claim 3d. *Let H be a subgroup of a group G , of index 2. Then H is normal.*

Proof. Let H have index 2 in a group G . Let $g \in G$, if $g \in H$ then $gH = H = Hg$ by closure. If $g \notin H$, since there are only two left cosets of H in G and $g \notin H$, they must be H and gH . Since the left cosets are pairwise disjoint but their union should be G , $gH = G \setminus H$. But the right cosets are also pairwise disjoint whose union is G , thus $Hg = G \setminus H$. Thus $gH = Hg$ for all $g \in G$ so H is normal. $\square$

Claim 3c. *Classify all subgroups of order of $|G| = 6$*

Proof. By Lagrange's theorem, elements of G must have order 1, 2, 3, 6.

1. Suppose G has order 6, then $G \cong \mu_6$.
2. Assume the contrary every element of G has order 2, then $G \cong F_2^n$. But $|G| = 6$ and $|F_2^n| = 2^n$ and there exists no natural number n such that $2^n = 6$. So G cannot have every element of order 2.
3. Assume the contrary every element had order 1, then each element would be the identity so the only element in G would be 1. Then order of G would be 1 and thus a contradiction would form and thus every element cannot have order 1 and thus a group like this wouldn't exist.
4. Assume the contrary every element had order 3. Let $g, h \in G$, suppose $\exists f \in \langle g \rangle \cap \langle h \rangle$, then $f = g^a$ and $f = h^b$ for $a, b \in [0, \dots, 2]$. If $a = b = 0$, then e would be in both subgroups. However if $a \neq 0$ and $b \neq 0$, thus $g^a = f^b$ but since 3 is prime and thus a coprime to 3, thus $g^a = f^b$ generates $\langle g \rangle$ but since b also coprime to 3, $g^a = f^b$ generates $\langle f \rangle$. Thus $f \in \langle g \rangle$ thus $\langle g \rangle = \langle f \rangle$. Thus the only element in the intersection of the cyclic subgroups with order 3 is e and all the other 2 elements in each subgroup were not in the intersection with any other cyclic subgroup. Thus the number of elements in G or its order is $2n + 1$ where n is the number of subgroups. $2n + 1 \equiv 1 \pmod{2}$. Thus $|G| \equiv 1 \pmod{2}$. This is a contradiction because $|G| \equiv 0 \pmod{2}$. Thus not all elements of G can be order 3.

Thus either G has order 6 or has element x of order 3 and element y of order 2.

If G is order 6, it is μ_6 as stated above. Otherwise, there exists x of order 3 and y of order 2. $[G : \langle x \rangle] = 2$, because the cosets would be split of elements in the cyclic group and those not thus $\langle x \rangle$ is the normal subgroup of G . Thus $\forall y \in G, yxy \in \langle x \rangle$. $yxy = x$ if $y \in \langle x \rangle$ and $yxy = -x$ if $y \notin \langle x \rangle$. Thus such a group would be congruent to S_3 . $\square$

Problem 9. *Double Cosets*

Proof. Let H, K be subgroups of G . Let $g, f \in G$. Let HgK and HfK be double cosets. Suppose $x \in HgK \cap HfK$ then $x = h_1 g k_1 = h_2 f k_2$. Thus $h_2^{-1} h_1 g k_1 k_2^{-1} = f$ and $h_2^{-1} h_1 \in H$ and $k_1 k_2^{-1} \in K$ by closure. Thus $f \in HgK$. Thus $HfK = HgK$. So the double cosets are pairwise disjoint and since $\forall g \in G \ ege = g \in HgK$ thus the union of all the double cosets is G and thus the double cosets partition G . $\square$

Problem 10. *Double cosets are the right coset iff H is normal.*

Proof. ($\implies$): Suppose double cosets are the right coset. Thus for all $g \in G$, $HgH = Hg$. But $egH \in HgH$ and but $egH = gH$ thus $gH \subset HgH$. Thus $gH = Hg$ and thus H is normal. ($\impliedby$): Suppose H is normal. Then $\forall g \in G$, and $\forall h_1, h_2 \in H$, $h_1 g h_2 \in HgH$ and $eh_1 g h_2 = gg^{-1} h_1 g h_2 = gh_3 h_2$ because H is normal $g^{-1} h_1 g = h_3 \in H$ thus $gh_3 h_2 \in gH$. Thus $HgH \subset gH$. $\forall gh \in gH$, $egh = gh \in HgH$, thus $gH \subset HgH$ thus $HgH = gH = Hg$ because H is normal. $\square$

Problem 12. *Postage Stamp Problem*

Proof. Since a and b are coprime, $[b]_a \in \mathbb{Z}/a\mathbb{Z}$ generates the group. Thus for all $k \in [0, \dots, a-1]$ $\exists n \in [1, \dots, a]$ s.t $nb \equiv k \pmod{a}$. Define $d_i := nb$ where $nb \equiv i \pmod{a}$. If $ab = ra + sb$ where $r, b \in \mathbb{N}$ either r, s must be 0, thus the minimum $n > ab$. $\forall m \geq ab + 1$, $m \cong j \pmod{a}$ for $j \in [0, \dots, a-1]$ then $m - d_j > 0$ because $d_j \leq ab$ thus $m - d_j \geq 1$. Furthermore $a \mid (m - d_j)$ because $d_j \equiv j \pmod{a}$, thus $ka = m - d_j$ thus $m = ka + d_j$ furthermore by definition $d_j = nb$ for $n > 0$ and $nb \equiv j \pmod{a}$. Thus $ka + nb = m$ for all $m \geq ab + 1$. $\square$